

a 3.5mm audio input port.

5V DC power supply (USB). You can use a 5V, 1A or 2A mobile charger.

FM antenna. It is not very critical. As MP3 player module does not come with antenna, you can use a whip antenna or a single-strand hook-up wire as antenna.

Wires and cables. The project requires a 3-wire cable for audio connection, a 2-wire cable for power supply, and a type-A USB cable. You also need a 3.5mm stereo audio jack with cable for audio output connection to audio amplifier.

Cardboard. You can use any thick/strong cardboard having a smooth surface for making the cabinet. For a sturdy cabinet, its size should be small. Cardboard DIY cabinets in various sizes are also available on the internet.

To make your own cardboard cabinet, draw the required size for each side of the cabinet on the cardboard using a scale and pencil/pen. After cutting according to the dimensions shown in Fig. 4, place the front, rear, and sides on a cardboard of 8.5cm × 14cm size. Fix each side on the base using glue gun.

As can be seen from the top view in Fig. 4, the cabinet has four sections or parts. On the left side, there are two 4.25cm × 2cm sections, which are to be used for holding the USBs. These two sections help in keeping the USBs in vertical position, so that they can be picked up easily when needed. The middle section is for holding MP3 player and the right one is for holding the IR remote. All these can be seen in the author's prototype (Fig. 1). Next, make two holes on the rear side for connecting the audio and power cables.

Fixing MP3 player module in the cabinet. Before fixing the MP3 player module into the cabinet, look out for the antenna point and solder it to the PCB. The internal part of MP3 player module is shown in Fig. 5.

Fixing the antenna. Use a suitable length (say, 10cm) of single-strand hook-up wire as antenna. The connecting point for the antenna in the MP3 module is not clearly visible. Hole for soldering the antenna is also not provided in the PCB. However, if you look carefully at the MP3 board on the other side, you can locate the antenna point for soldering, as shown in Fig. 6. You can solder the antenna wire here using a 25W soldering iron. All soldering work should be done before fixing the MP3 module in the cabinet.

Soldering the cables. Before connecting the audio and power cables in the MP3 player module, you need to connect the audio cable to an audio

YOU CAN NOW SET UP AN EFY-CERTIFIED SKILL DEVELOPMENT CENTRE IN YOUR INSTITUTION

In Partnership with EFY



EFY-CERTIFIED
Skill DEVELOPMENT
CENTRE

Referred to as ESDC, the Centre will help students improve their hands-on skills, make exciting projects of their own and understand the latest technologies

For more details, please call: Shyam (+91-8800100761)
email: efyenq@efy.in or visit: <https://efymag.com/ESDC>